



TavaOne Education

General License Course · 2023–2027 Pool

Module 3

Equipment, Practices & Electronics

General License Course · G4 + G5 + G6 · 10 Exam Questions



Hour 1

Station Equipment and Practices



G4A — Transceiver Controls You Need to Know

RF Gain: controls the sensitivity of the receiver front end — turn it down to reduce overload from strong signals or high noise

AF Gain: volume control — adjusts audio output level only

Squelch: mutes audio when no signal present — rarely used on HF (too much noise), standard on VHF FM

Noise Blanker (NB): removes impulse noise (ignition noise, power line hash) by detecting brief high-amplitude pulses and blanking the receiver

Notch Filter: eliminates a specific interfering carrier — tune the notch to the exact frequency of the interferer

Passband Tuning (PBT): shifts the IF passband to move interference out of the audio; different from notch

VOX: voice-operated transmit; microphone audio triggers PTT automatically; set the threshold carefully

Speech Processor: increases average power by compressing audio peaks — sounds louder to the other station



G4A — TR Sequencing and External Amplifiers

External linear amplifier: amplifies the transceiver's output, typically 100W → 500W–1500W

Problem: if you key the amplifier before the transceiver switches from receive to transmit, you hot-switch the PA

TR Sequencing: the order in which the system switches from receive to transmit

Correct sequence: 1) PTT pressed → 2) transceiver goes to transmit mode → 3) relay switches → 4) amplifier is keyed

Built-in delay circuits (or separate sequencers) handle this; most modern transceivers have a key delay setting

Hot-switching the PA can destroy expensive amplifier tubes or transistors

CW: semi-breakin (QSK semi) vs. full breakin (QSK) — full breakin hears between dit spaces, slow CW only

VOX delay: prevents the radio from dropping out between words; too short = choppy, too long = can't receive replies promptly



G4B — Test Equipment

Oscilloscope: displays voltage waveforms vs. time; use to check modulation, look for key clicks, check power supply ripple

Directional Wattmeter (SWR meter): measures forward and reflected power; placed in the feed line between transceiver and antenna

Dummy Load: 50-ohm resistive load (no radiation); use when testing or tuning to avoid transmitting on the air unnecessarily

Antenna Analyzer: measures feed-point impedance, SWR, resonant frequency; invaluable for antenna work

Frequency Counter: measures transmitted frequency precisely; useful for VFO calibration

Signal Generator: creates a test signal at a known frequency and level; use to test receiver sensitivity

Multimeter (DMM): measures DC voltage, current, resistance, continuity; essential for troubleshooting

Spectrum Analyzer: displays power vs. frequency; shows harmonics, spurious emissions (expensive)



G4C — RF Interference and Remedies

Common-mode current on coax shield: RF flows back along the outside of the coax into the shack, gets into audio equipment

Solution: choke balun (ferrite core) at the antenna feedpoint; snap-on ferrites on cables

TVI (television interference): usually harmonics from the transmitter; most TVs have improved rejection but still possible

Audio equipment interference: RF gets rectified by op-amp input diodes and appears as audio distortion

Fix for audio interference: ferrite choke on the microphone cable at the radio; snap-on cores work well

Telephone interference: RF on phone lines; ferrite chokes at the phone handset

BCI (broadcast interference to your receiver): a strong AM station mixing in your receiver front end

Your responsibility: you must not cause harmful interference to other services; neighbors' interference is often also your issue to resolve



G4D — SSB Microphone Technique and Dynamic Range

SSB mic technique: speak across the mic, not into it; keep distance consistent; avoid breathing directly at the mic

Audio compression: speech processor compresses dynamic range → higher average power → louder on the other end (but sounds more compressed)

Receiver dynamic range: ability to receive a weak signal in the presence of a strong one on a nearby frequency

Intermodulation distortion (IMD): two strong signals mixing in the receiver front-end create a third (unwanted) signal

IMD products: $f = n \times f_1 \pm m \times f_2$; third-order products ($2f_1 - f_2$) fall close to the original signals and are the most problematic

Desensitization: very strong signal overloads the front end, reducing sensitivity to weaker signals

Solution to IMD/desens: reduce RF gain, add attenuation, use a bandpass filter before the receiver

Mobile HF noise: alternator whine (RPM-correlated), ignition noise (pops); fix with ferrite chokes, ground straps, noise filter on power leads



Hour 2

Electronics: Reactance, Components, and Circuits



G5A — Impedance and Reactance

Impedance (Z): total opposition to AC current flow = combination of resistance (R) and reactance (X)

Reactance: opposition to current caused by capacitors or inductors; changes with frequency

Capacitive reactance: $X_c = 1/(2\pi fC)$ — as frequency INCREASES, X_c DECREASES; capacitor acts like a short at high frequency

Inductive reactance: $X_L = 2\pi fL$ — as frequency INCREASES, X_L INCREASES; inductor acts like an open at high frequency

At resonance: $X_L = X_c$; net reactance = 0; impedance = purely resistive

Series resonance: impedance is MINIMUM at resonance ($Z = R$ only)

Parallel resonance: impedance is MAXIMUM at resonance (very high Z at resonant frequency)

50 ohms: why all amateur radio gear is 50 ohms — it's a compromise between minimum loss and maximum power transfer in coax



G5B and G5C — Circuit Calculations

Power factor: ratio of real power (watts) to apparent power (VA); $PF = \cos(\text{phase angle})$; unity PF = purely resistive

Series resistors: $R_{\text{total}} = R1 + R2 + \dots$ (just add them)

Parallel resistors: $1/R_{\text{total}} = 1/R1 + 1/R2 + \dots$; two equal resistors: half the value

Series capacitors: $1/C_{\text{total}} = 1/C1 + 1/C2$ (like parallel resistors — they get smaller)

Parallel capacitors: $C_{\text{total}} = C1 + C2$ (just add them)

Series inductors: $L_{\text{total}} = L1 + L2$ (just add them)

Parallel inductors: $1/L_{\text{total}} = 1/L1 + 1/L2$ (like parallel resistors)

Transformer turns ratio: $V_s/V_p = N_s/N_p$; impedance ratio = $(\text{turns ratio})^2$; 2:1 turns = 4:1 impedance



G6A & G6B — Semiconductor Components

Diodes

Rectifier diode: allows current in one direction only; converts AC to pulsating DC

Zener diode: operates in reverse breakdown at a specific voltage; used for voltage regulation

Varactor diode: capacitance varies with reverse voltage; used in VFOs and PLLs for electronic tuning

LED: light-emitting diode; emits photons when forward biased

Transistors and ICs

NPN bipolar: base current controls collector-emitter current; current flows collector to emitter when base is +

PNP bipolar: opposite polarity; less common in RF work

MOSFET: voltage at gate controls drain-source current; very high input impedance; common in RF amplifiers

Op-amp: differential input, very high gain; use feedback to set closed-loop gain

Crystal: piezoelectric resonator; extremely stable frequency; used in oscillators and filters



G6B — The Phase-Locked Loop (PLL)

PLL: Phase-Locked Loop — the heart of every modern synthesized transceiver

Purpose: generate a precise, stable frequency by comparing a divided VCO output to a crystal reference

How it works: VCO → frequency divider → phase detector → low-pass filter → back to VCO control voltage

The VCO (voltage-controlled oscillator) changes frequency until its divided-down output matches the reference crystal

Change the division ratio → change the output frequency → this is how digital tuning works

PLLs are fast (frequency changes in microseconds) and stable (locked to the crystal)

Phase noise: a PLL's limitation — the output isn't perfectly clean; sidebands appear near the carrier

Practical: every time you tune your Icom or Yaesu, the PLL is setting the new frequency



Checkpoint Quiz

What is the purpose of a noise blanker in a receiver?

A. To reduce or eliminate pulse-type interference such as ignition noise

B. To reduce random static from atmospheric noise

C. To eliminate a specific interfering carrier signal



Checkpoint Quiz

What is the relationship between capacitive reactance and frequency?

A. Capacitive reactance decreases as frequency increases

B. Capacitive reactance increases as frequency increases

C. Capacitive reactance is constant regardless of frequency



Checkpoint Quiz

What type of diode is commonly used for electronic tuning in receivers and transmitters?

A. Varactor diode

B. Zener diode

C. Schottky diode



Checkpoint Quiz

Two resistors of 10 ohms each are connected in parallel. What is the total resistance?

A. 5 ohms

B. 20 ohms

C. 10 ohms



Checkpoint Quiz

What is the function of a speech processor in an SSB transmitter?

A. To increase average power by reducing the peak-to-average ratio of the voice signal

B. To improve audio quality by filtering unwanted frequencies

C. To increase transmitter bandwidth for clearer audio



HamStudy Assignment

Drill these sections before next module:

G4A — Station Operation & Controls

G4B — Test Equipment

G4C — RF Interference

G4D — SSB & Dynamic Range

G4E — Mobile & Field Ops

G5A — Impedance & Reactance

G5B/C — Circuit Calculations

G6A — Diodes

G6B — Transistors & PLLs

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